

Chapter 104 — Pathophysiology and Classification of LUT Dysfunction

IF YOU READ NOTHING ELSE

- All LUT dysfunction = bladder or outlet, failing to store or empty (Wein)
- Enhorning: stress continence needs $\geq$ equal pressure transmission to proximal urethra
- UUI $\rightarrow$ assume detrusor overactivity (DO); urgency alone often no DO on UDS
- BOO hallmark: high detrusor pressure + low flow
- Polyuria >40 mL/kg/24 h (≈ 2.8 L at 70 kg)
- Nocturnal polyuria: NPI $>20\%$ young, $>33\%$ elderly
- ISD alone can cause SUI; hypermobility alone cannot
- Suprasacral $\rightarrow$ DO $\pm$ DSD; sacral $\rightarrow$ acontractile; suprapontine $\rightarrow$ synergic
- Male sphincteric UI = female ISD, never hypermobility
- Collagen-driven low compliance $\rightarrow$ drug-unresponsive

NORMAL LUT FUNCTION

- Storage passive: elastic/viscoelastic wall, no parasympathetic drive
- Voiding centre: rostral brainstem; parasympathetic outflow sacral (pelvic nerve)
- Guarding reflex: pudendal $\rightarrow$ $\uparrow$ striated sphincter activity on filling
- $\alpha_1 \rightarrow \uparrow$ bladder neck tone; $\beta_3 \rightarrow$ detrusor relaxation
- Smooth sphincter: involuntary, physiological not anatomical
- Extramural striated sphincter = voluntary external sphincter (male membranous, female mid-urethra)
- Bladder neck relaxation: NANC (non-adrenergic non-cholinergic), probably NO
- Funnelling: base-urethra longitudinal muscle continuity
- Most important afferents: pelvic nerve $\rightarrow$ sacral cord
- Normal filling sensation: fullness — never urgency or pain
- Klevmark: cystometric pressure rise = supraphysiological filling artefact
- Tanagho (1978): active reflex rise in closure pressure on straining
- Mucosal seal: atrophic/scarred mucosa seals poorly
- Oestrogen for postmenopausal UI: controversial

PATHOPHYSIOLOGY

- Storage failure: bladder overactivity (DO or $\downarrow$ compliance), $\downarrow$ outlet resistance, altered sensation
- Emptying failure: $\downarrow$ contractility or $\uparrow$ outlet resistance
- OAB (ICS): urgency $\pm$ UUI, usually frequency + nocturia
- Low compliance: sacral/intrasacral neurology; long-standing BOO (collagen)
- Genuine SUI: competent at rest, fails on strain; ISD: incompetent at rest
- ISD overt: open bladder neck, low ALPP and closure pressure; after surgery, scarring, neurology
- ISD $\rightarrow$ coaptation/compression; hypermobility-only repair fails
- DeLancey hammock: unstable suburethral layer $\rightarrow$ SUI
- Continence point in $\sim$ half of continent females: mid-urethra
- Hypermobility alone $\neq$ diagnosis: needs demonstrated UI
- Underactivity causes: neurogenic, iatrogenic pelvic surgery, overdistension, drugs, diabetes
- ICS BOO: $\downarrow$ flow and/or $\uparrow$ PVR + $\uparrow$ detrusor pressure; commoner in males
- Male BOO: prostate, bladder neck contracture, stricture
- Female BOO: prolapse, diverticulum, post-incontinence-surgery
- Striated sphincter dyssynergia: common functional obstruction (neurological)
- True smooth sphincter dyssynergia: unusual, except autonomic dysreflexia

- Bladder neck dysfunction: young/middle-aged, normal anatomy
- Obstruction $\rightarrow$ intervene promptly: hypertrophy, collagen, diverticula, DO
- Emptying failure fallback: intermittent catheterisation
- Flow + PVR integrate bladder and outlet; DLPP bladder, VLPP outlet
- Storage drugs: antimuscarinics, β -agonists, onabotulinumtoxinA (intradetrusor); outlet: α -agonists
- Striated sphincter relaxants: baclofen, benzodiazepines, dantrolene, botulinum

PAIN, NOCTURIA, NLUTD

- IC/BPS urgency: driven by pain worsening with filling
- A δ = myelinated, normal; C = unmyelinated, silent, sensitised
- Central sensitisation $\rightarrow$ OAB-IBS crosstalk
- NPI = night volume (incl. first morning void) $\div$ 24-h
- Polyuria causes: DM, central/nephrogenic DI, primary polydipsia
- Elderly: lost nocturnal vasopressin rise
- OSA $\rightarrow$ $\uparrow$ ANP $\rightarrow$ $\downarrow$ vasopressin $\rightarrow$ nocturnal polyuria
- NLUTD (ICS): needs confirmed relevant neurological disorder

CLASSIFICATION SYSTEMS

- Four: function, symptoms, signs, urodynamics
- Functional: no exact diagnosis or UDS needed to treat
- Symptoms cannot alone diagnose
- T10 paraplegia: DO + dyssynergia; treat one as primary
- T10 UDS: neurogenic DO, absent sensation, low capacity, DSD
- Stroke UUI: neurogenic DO, normal sensation, normal voiding
- Cystometric capacity: "permission to void"; maximum: cannot delay
- Underactive = inadequate magnitude/duration; acontractile = none seen
- Normal urethral storage: positive closure pressure

CLASSIC TRAPS

- Trap: raised intravesical pressure = voiding — Actually: cough needs no contraction
- Trap: hypermobility = SUI — Actually: common in continent females
- Trap: urgency without leak excludes DO — Actually: suggests DO
- Trap: low flow = obstruction — Actually: flow integrates bladder and outlet
- Trap: underactivity only an emptying failure — Actually: also overflow storage failure
- Trap: DSD from suprapontine lesion — Actually: suprasacral; suprapontine synergic
- Trap: terminal dribble post-micturition — Actually: voiding; after-dribble is post-micturition

- **Trap:** C fibres large-calibre — **Actually:** small, unmyelinated

- **Trap:** NPI excludes first morning void — **Actually:** includes it
- **Trap:** nocturia needs ≥ 2 wakings — **Actually:** one or more